A2 · Unit 5 · WBI15 Respiration, Internal Environment, Coordination and Gene Technology

Muscles and the Heart

Movement depends on muscles pulling on bones through tendons, and muscle contraction depends on actin and myosin sliding past each other. The heart is a muscle too, but it contracts on its own rhythm. This page covers the skeleton and muscle structure, the sliding filament mechanism, fibre types, and how the heart's electrical activity produces the heartbeat and an ECG.

SPEC 7.9 Muscles, tendons, ligaments and the skeleton

WHAT THIS SPEC POINT ASKS

Know how muscles, tendons, the skeleton and ligaments interact to produce movement, including antagonistic muscle pairs, extensors and flexors.

The parts

Structure	Joins	Properties	Job
Bone	–	rigid and strong (collagen + calcium phosphate)	lever; support; attachment for muscles
Tendon	**muscle to bone**	strong, **inelastic** (mostly collagen)	transmits the pull of the muscle to the bone without losing force
Ligament	**bone to bone**	strong, **slightly elastic** (collagen + more elastic fibres)	holds the bones of a joint together, keeps it **stable**, allows some movement
Cartilage	covers the ends of bones	smooth, tough, flexible	reduces friction, absorbs shock
Synovial fluid	fills the joint cavity	–	lubricates the joint

Antagonistic pairs

- Muscles can only **pull** (contract). They can't push. So each movable joint needs two muscles that pull in opposite directions: an **antagonistic pair**.
- A **flexor** bends (flexes) a joint. An **extensor** straightens (extends) it.

- When one contracts, its antagonist **relaxes** and is **stretched** back to its original length by the contracting muscle.
- Elbow: **biceps** = flexor; **triceps** = extensor. Knee: hamstrings = flexors; quadriceps = extensors.
- To hold a joint **steady** at a fixed angle, **both** muscles contract with equal force, and ligaments hold the bones in place.

WORKED EXAMPLE

Worked example: describe how the arm is straightened (2 marks)

The **triceps (extensor) contracts** and pulls on the lower arm bone (ulna) through its tendon.

The **biceps (flexor) relaxes** and is stretched. The arm straightens at the elbow.

MARK-SCHEME WORDING

- Ligament: "joins **bone to bone**", "maintains joint **stability**". Tendon: "joins **muscle to bone**", "**transmits force**".
- "Flexor and extensor muscles are **antagonistic**"; "one **contracts** while the other **relaxes**". The words "joint" and "straightening" from the stem are ignored.
- Name **both** muscles of the pair: naming only one scored 0 (WBI15 Jun 2024).
- Composition: both are mostly **collagen** (strength); ligaments contain more **elastic** fibres so the joint can stretch a little; tendons stay stiff so force isn't lost.
- Ligament damage needs a stated cause (a twisting injury, overextension, a tackle). "Injury" alone gains nothing.

Often asked as: MCQs on tendons and ligaments; label the flexor, extensor and ligament on an elbow diagram; describe how the arm is straightened; how the knee is held steady; tendon and ligament composition and stiffness data.

COMMON MISTAKE

- Swapping tendons and ligaments, or flexors and extensors (labelling the triceps as a flexor).
- Writing about biceps and triceps when the question is about the knee.
- "The muscle pushes the bone back." Muscles only pull: the antagonist makes the return movement.
- Stopping at "holds bones together" without adding "keeps the joint **stable**".

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- Collagen and elastin percentages and stress–strain slopes of tendons and ligaments, e.g. a slope ratio of 11.25 : 1 (WBI15 Oct 2025).

- Tendons get less stiff with age, so less force reaches the bone, contributing to falls in older people (WBI15 Jun 2021 article).
- Skeletal muscle is **neurogenic** (it contracts only when stimulated by motor neurones), not myogenic (WBI15 Jan 2022).

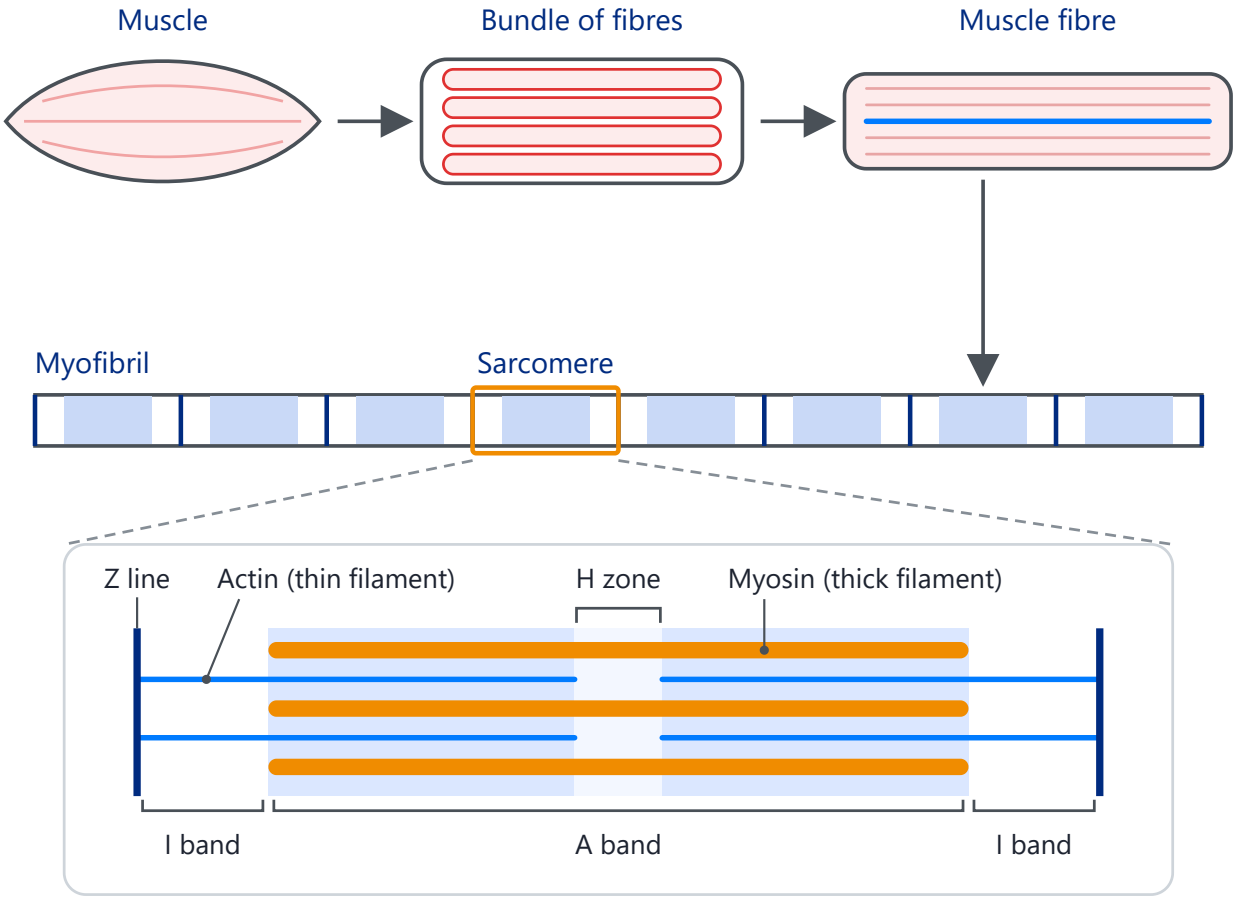
SPEC 7.10 Skeletal muscle fibres; fast and slow twitch

WHAT THIS SPEC POINT ASKS

(i) Know the structure of a mammalian skeletal muscle fibre. (ii) Understand the structural and physiological differences between fast-twitch and slow-twitch muscle fibres.

From muscle to filament

muscle → bundles of **muscle fibres** (the cells) → each fibre packed with **myofibrils** → each myofibril is a chain of **sarcomeres** → made of **actin** (thin) and **myosin** (thick) filaments.



From a whole muscle down to the sarcomere, where the thin actin and thick myosin filaments overlap.

The muscle fibre

Part	What it is	Why it matters
Multinucleate fibre	one long cell formed when many cells fuse	many nuclei to make proteins along its length; no membranes between sections, so the whole fibre contracts together
Sarcolemma	the cell surface membrane	carries the action potential
T-tubules (transverse tubules)	infoldings of the sarcolemma running deep into the fibre	carry the action potential into the fibre, close to the sarcoplasmic reticulum
Sarcoplasm	the cytoplasm	contains glycogen and myoglobin
Sarcoplasmic reticulum (SR)	specialised endoplasmic reticulum wrapped around each myofibril	stores **Ca^{2+}**, releases it when stimulated, pumps it back afterwards
Many mitochondria	packed between the myofibrils	ATP for contraction and for pumping Ca^{2+}
Myofibrils	bundles of actin and myosin filaments	the contractile units

The sarcomere and its bands

- A **sarcomere** is the repeating unit between two **Z lines** (Z discs).
- **A band** (dark): the length of the **myosin** filaments, including where they overlap with actin.
- **I band** (light): **actin only**. Each I band spans a Z line, so half of it lies in each neighbouring sarcomere.
- **H zone**: **myosin only**, in the middle of the A band.
- **M line**: the centre of the sarcomere, where the myosin filaments are held together.
- The alternating dark and light bands make skeletal muscle look **striated** (striped). See the figure in 7.11.

Fast-twitch v slow-twitch fibres

	Slow twitch (type I)	Fast twitch (type II)
Contraction	slower, less powerful, **sustained**	faster, more powerful, short bursts
Fatigue	resistant	fatigue quickly

	Slow twitch (type I)	Fast twitch (type II)
Main ATP source	**aerobic** respiration	**anaerobic** respiration and creatine phosphate
Mitochondria	**many**	fewer
Myoglobin	**lots** (dark red)	little (pale)
Capillaries	**dense** network	fewer
Glycogen and creatine phosphate	less	**more**
Sarcoplasmic reticulum	less	**more** (fast Ca^{2+} release)
Fibre diameter	smaller (short diffusion distance for O_2)	larger
Examples	posture muscles in the back; marathon runners' legs; chicken leg	sprinters' legs; eyelid muscles; chicken breast

- **Myoglobin** is a red, oxygen-binding protein in muscle. It has a **higher affinity for O_2** than haemoglobin, so it takes O_2 from the blood and **stores** it, releasing it only when the muscle's O_2 level is very low.

EXAM TIP

"Explain why a marathon runner needs slow-twitch fibres" needs **comparatives** ("more mitochondria", "more capillaries") **and** a link to the athlete: many mitochondria → more aerobic respiration → more ATP over a long time without lactate building up → less fatigue.

WORKED EXAMPLE

Worked example: sarcomere length from an electron micrograph

Three sarcomeres, measured from the first Z line to the fourth, are **66 mm** long on a micrograph at × 10 000.

One sarcomere on the image = 66 ÷ 3 = 22 mm = 22 000 µm.

Actual length = image ÷ magnification = 22 000 ÷ 10 000 = **2.2 µm**.

Always measure from Z line to Z line, and convert the units before dividing.

MARK-SCHEME WORDING

- Fibre structure: "single **multinucleate** cell", "repeating bands / sarcomeres", plus sarcolemma, SR, T-tubules, many mitochondria.
- Fast twitch compared with slow twitch (comparatives essential; converse accepted): "**fewer** mitochondria", "**less** myoglobin", "**fewer** capillaries", "**larger** diameter", "**more** sarcoplasmic reticulum", "**more** glycogen", "**more** creatine phosphate". "Less blood supply" alone is ignored because it isn't a structure.
- Slow-twitch benefits: "many mitochondria give more **aerobic** respiration / ATP"; "myoglobin **stores** oxygen"; "large capillary network supplies O_2 and glucose"; "slow, sustained contraction, so **less fatigue**".
- Sprinter: "fast speed over a short time"; "fast-twitch fibres **contract more quickly**"; "rely on **anaerobic** respiration or **creatine phosphate**, not on oxygen supply".
- Sarcomere: "the repeating unit **between Z lines**". Type 2 / IIa / IIb / IIx are accepted for fast twitch.

Often asked as: describe a muscle fibre; compare fast- and slow-twitch fibres; why sprinters or marathon runners have more of one type; the role of myoglobin; sarcomere length, ratios and magnification from an electron micrograph; training and fibre type.

COMMON MISTAKE

- Slow-twitch fibres have capillaries "to store O_2". Capillaries **deliver** O_2; **myoglobin stores** it.
- Listing properties without comparatives, or without linking a feature to what the athlete needs.
- Giving function when the question asks for **structure** ("fatigue quickly" is not a structure).
- "More mitochondria" the wrong way round (slow twitch has more).
- Putting units on a magnification, or rounding it wrongly (15 500 is not 16 000).

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- **Intermediate** (type IIa) fibres: endurance training converts some fast fibres into intermediate fibres; slow fibres don't change; more training gives a bigger switch (WBI15 Jan 2021).
- **Creatine phosphate** in fast-twitch fibres regenerates ATP quickly by giving its phosphate to ADP (WBI15 Jun 2021 article).
- **Sarcopenia** (age-related loss of muscle), satellite cells, loss of motor units and percentage-change calculations (WBI15 Jun 2021, Jan 2024 articles).
- Ratio of relaxed to contracted sarcomere length from a micrograph, e.g. 1.8 : 1 (WBI15 Jun 2024).

SPEC 7.11 Sliding filament theory

WHAT THIS SPEC POINT ASKS

Understand how skeletal muscle contracts according to the sliding filament theory, including the roles of actin, myosin, troponin, tropomyosin, calcium ions (Ca^{2+}), ATP and ATPase.

The proteins

- **Myosin** (thick filament): long tails with **heads** that stick out. Each head can bind to actin and has **ATPase** activity (it hydrolyses ATP).
- **Actin** (thin filament): has **binding sites** for myosin heads.
- **Tropomyosin**: a long, thin protein that winds along the actin filament and **covers the myosin-binding sites** when the muscle is relaxed.
- **Troponin**: a protein attached to tropomyosin that can **bind Ca^{2+}**.

Contraction, step by step

1. An impulse arrives at the **neuromuscular junction**. Acetylcholine causes the **sarcolemma** to depolarise, and the action potential spreads along it and down the **T-tubules**.
2. This makes the **sarcoplasmic reticulum release Ca^{2+}** into the sarcoplasm.
3. Ca^{2+} **binds to troponin**, which **changes shape** and **pulls tropomyosin** away from the binding sites. The **myosin-binding sites on actin are exposed**.
4. A **myosin head** (already carrying ADP + Pi) **binds** to a site on actin, forming a **cross-bridge**.
5. The head **tilts** (the **power stroke**), releasing ADP + Pi and **pulling the actin filament towards the M line** (the centre of the sarcomere).
6. A new **ATP binds** to the myosin head. This makes the head **detach** from actin, breaking the cross-bridge.
7. **ATPase** on the myosin head **hydrolyses the ATP** to ADP + Pi. The energy released **returns the head to its upright ("cocked") position**.
8. The head binds to a new site further along the actin, and the cycle repeats as long as Ca^{2+} and ATP are present. Thousands of heads cycle out of step, so the pull is smooth.

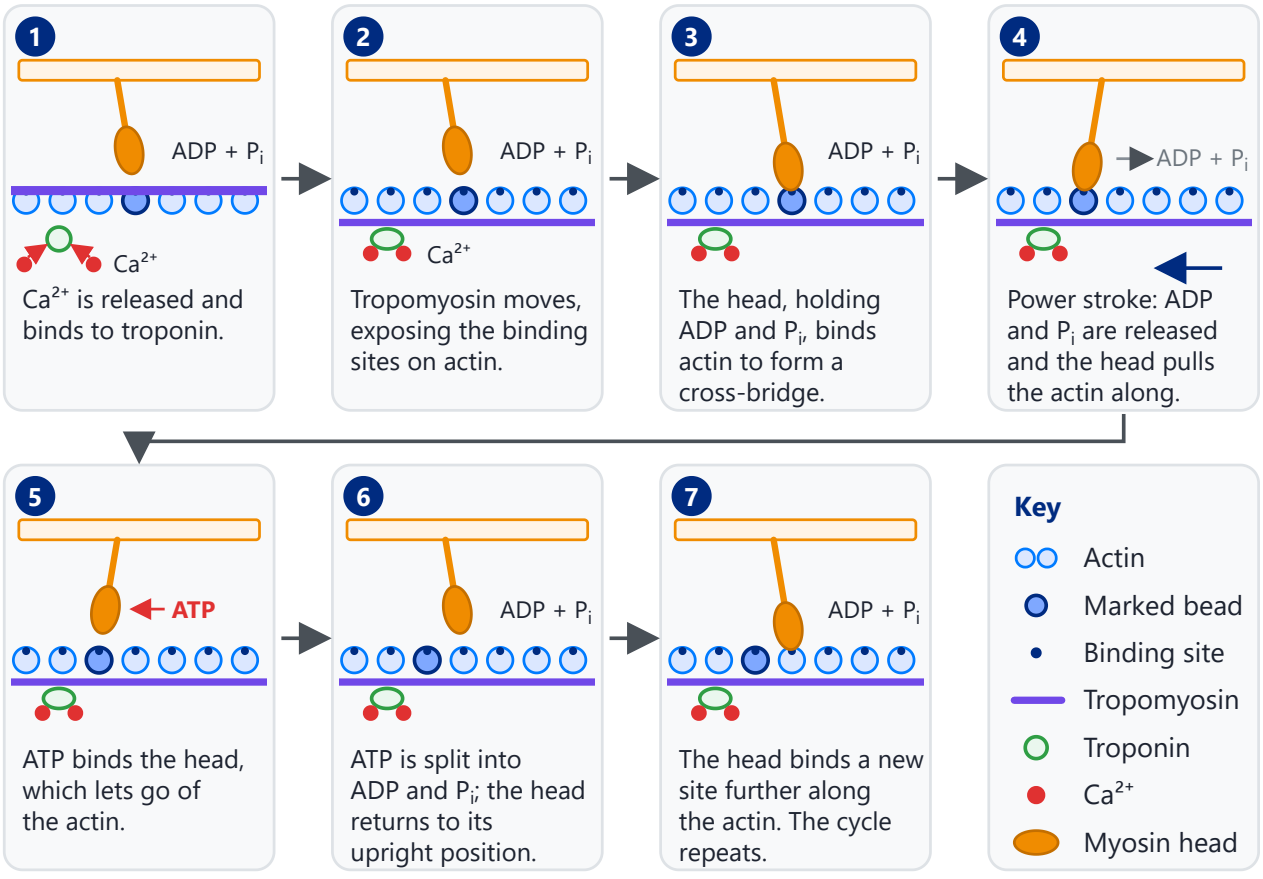
One cycle of cross-bridge movement

1. A nerve impulse reaches the neuromuscular junction, and calcium ions are released from the sarcoplasmic reticulum.
2. Calcium binds troponin, which moves tropomyosin away from the binding sites on actin.

   Without this step, the myosin heads cannot reach actin, so no contraction can start.

3. The myosin head, already carrying ADP and phosphate from the previous cycle, binds to actin and forms a cross-bridge.
4. ADP and phosphate are released, and the head swings forwards (the power stroke), pulling the actin filament along.
5. A fresh ATP molecule binds the head and the bond with actin breaks.
6. The ATPase splits ATP into ADP and phosphate, and the head returns to its upright position, storing energy.

The energy for the return stroke comes from ATP hydrolysis, so the head can attach again further along the actin.



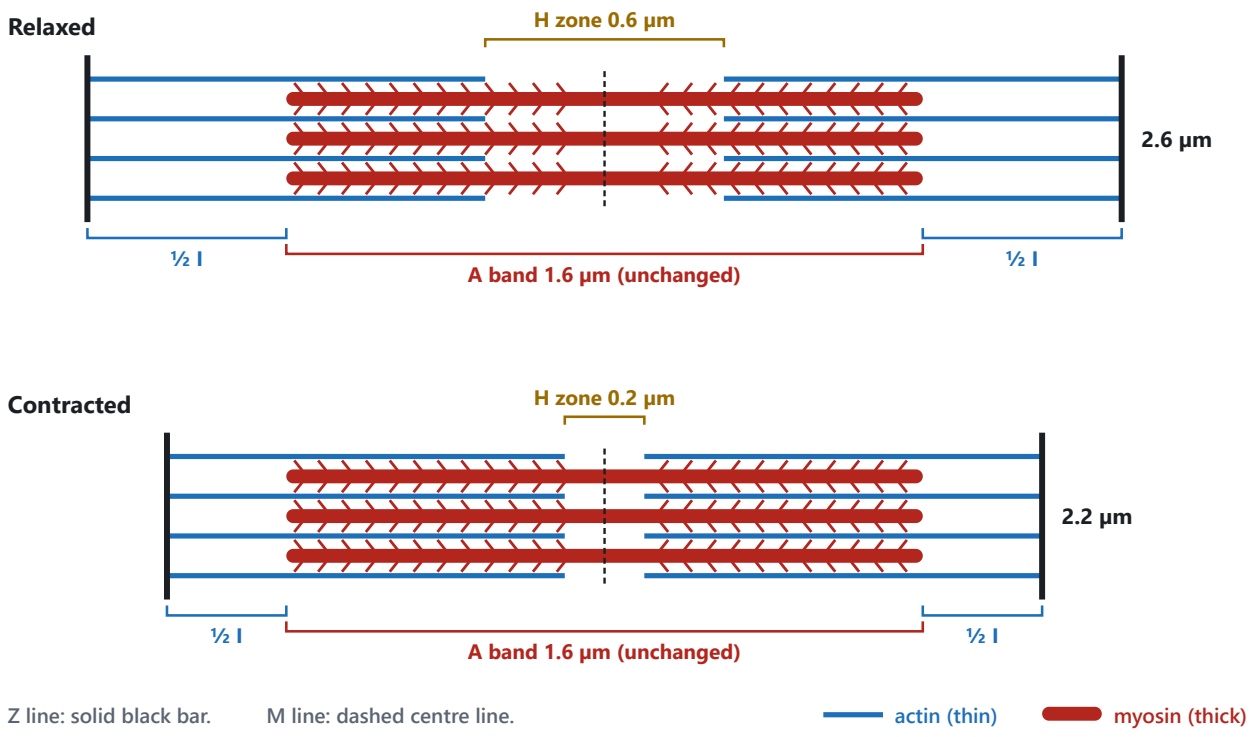
The seven stages of cross-bridge cycling, from unblocking the binding site to reattaching further along the actin filament.

Relaxation

- When impulses stop, Ca^{2+} is **actively transported back into the SR** (this uses ATP).

- Troponin loses its Ca^{2+} and tropomyosin moves back to **cover the binding sites**. No cross-bridges can form, so the muscle relaxes and is pulled back to its full length by its antagonist.

What changes length



One sarcomere, relaxed (2.6 µm) and contracted (2.2 µm). The actin and myosin filaments keep their length and slide past each other: the sarcomere, the I band and the H zone shorten; the A band stays the same.

	On contraction
Sarcomere (Z line to Z line)	**shorter**
I band	**shorter**
H zone	**shorter** (may disappear)
A band	**no change** (it is the length of the myosin)
Actin and myosin filaments	**no change**: they slide past each other

The first interactive diagram at the foot of the page lets you contract a sarcomere yourself.

WORKED EXAMPLE

Worked example: band lengths

In the figure, the myosin filaments are 1.6 μm long and each actin filament reaches 1.0 μm from its Z line.

Relaxed sarcomere, 2.6 μm: I band (two halves) = 2.6 − 1.6 = **1.0 μm**; H zone = 2.6 − (2 × 1.0) = **0.6 μm**.

Contracted sarcomere, 2.2 μm: I band = 2.2 − 1.6 = **0.6 μm**; H zone = 2.2 − 2.0 = **0.2 μm**.

A band = **1.6 μm** in both. Ratio of relaxed to contracted length = 2.6 ÷ 2.2 = **1.2 : 1** (1 d.p.).

Two jobs for ATP

- **Detaching** the myosin head from actin (when ATP binds) and **re-cocking** it (when ATPase hydrolyses ATP).
- **Pumping Ca^{2+}** back into the SR so the muscle can relax.
- **Rigor mortis**: after death no ATP is made, so myosin heads can't detach and the muscles lock rigid.
- **Creatine phosphate** in muscle remakes ATP quickly: creatine phosphate + ADP → creatine + ATP.

MARK-SCHEME WORDING

- Calcium route: "Ca^{2+} **released from the sarcoplasmic reticulum**"; "**binds to troponin**, which **changes shape**"; "**tropomyosin moves**"; "**exposes** (myosin) **binding sites on actin**"; "myosin head binds, forming a **cross-bridge**".
- ATP route, in logical order: "ADP and Pi released, myosin head **pulls actin** towards the centre of the sarcomere"; "**ATP binds** to the myosin head and **breaks the cross-bridge**"; "ATP **hydrolysed** (by ATPase on myosin), providing energy to **return the head** to its original position"; "head can bind again".
- Say "**myosin head**", not just "myosin". "The power stroke" is **not** accepted for the ATP-driven recovery step.
- Less Ca^{2+}: less binds to troponin, tropomyosin stays in place, **fewer cross-bridges**, the muscle stays relaxed or contracts weakly.
- Force against velocity: faster shortening means fewer cross-bridges attached at any moment, so **less force** ("inversely proportional" or "negative correlation" accepted).

Often asked as: describe the sliding filament theory (4 marks); the role of Ca^{2+}, troponin or ATP on its own; which part of the sarcomere shortens; why less Ca^{2+} disrupts contraction; isolated fibres in ATP or glucose solutions.

COMMON MISTAKE

- Mixing up the power stroke (head tilts, ADP + Pi released) with the recovery stroke (ATP hydrolysed, head returns upright). This was the commonest error in the examiner reports.
- Saying the actin or myosin filaments **shorten**. They slide; the sarcomere shortens.

- Saying Ca^{2+} binds to tropomyosin, or that tropomyosin "frees the myosin heads". Ca^{2+} binds **troponin**, and tropomyosin uncovers the binding sites **on actin**.
- Writing out the whole theory when the question asks only about ATP, or only about troponin.
- "Myoglobin head." It's the **myosin** head.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- Isolated muscle fibres: in an **ATP** solution they shorten straight away (heads can detach and re-cock); in **glucose** they shorten less, because ATP must first be made by respiration. The solutions had the same water potential as the cytoplasm so the fibres didn't change length by osmosis (WBI15 Jan 2026).
- Creatine phosphate as a fast ATP regenerator (WBI15 Jun 2021 article).
- T-tubules and the Ca^{2+} release channels of the SR (ryanodine receptors) in an article context (WBI15 Jun 2021).

SPEC 7.12 The myogenic heart, its conduction system and ECGs

WHAT THIS SPEC POINT ASKS

(i) Know that cardiac muscle is myogenic. (ii) Understand how the heart's own electrical activity coordinates the heartbeat: the sinoatrial node (SAN), atrioventricular node (AVN), bundle of His and Purkyne fibres. (iii) Understand how electrocardiograms (ECGs) help diagnose abnormal heart rhythms.

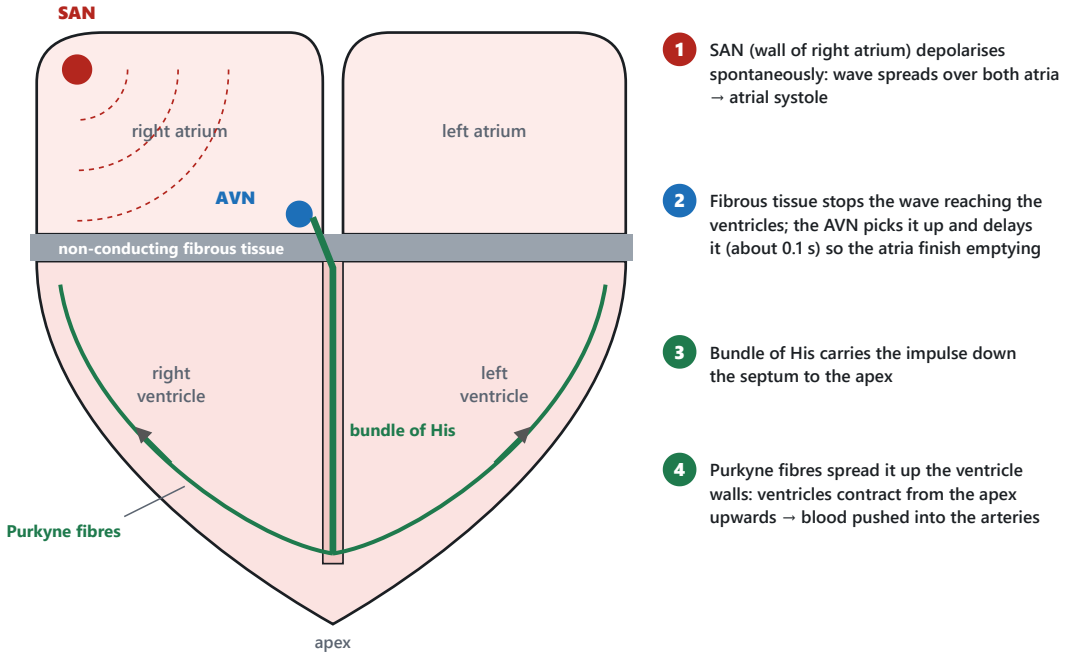
Myogenic

- **Myogenic**: cardiac muscle starts each contraction **itself**, without nerve impulses. Nerves and hormones only **change the rate** (7.13).
- Skeletal muscle is **neurogenic**: it contracts only when a motor neurone stimulates it.

How the heartbeat is coordinated

1. The **sinoatrial node (SAN)**, a patch of specialised cardiac muscle in the wall of the **right atrium**, depolarises spontaneously and regularly. It is the **pacemaker**.
2. A **wave of excitation** (depolarisation) spreads across both **atria**, which contract together (**atrial systole**), pushing blood into the ventricles.
3. A layer of **non-conducting fibrous tissue** between the atria and ventricles stops the wave spreading straight into the ventricles.
4. The **atrioventricular node (AVN)** picks up the wave and **delays it** by about 0.1 s. This gives the atria time to **finish contracting and emptying** before the ventricles contract.
5. The AVN passes the excitation down the **bundle of His**, in the **septum**, to the **apex** at the bottom of the ventricles.

6. **Purkyne fibres** carry it up through the ventricle walls. The ventricles contract **from the apex upwards** (**ventricular systole**), squeezing blood up and out into the aorta and pulmonary artery.
7. The muscle repolarises and relaxes (diastole) until the SAN fires again.



The conduction pathway (schematic; the right side of the heart is drawn on the left). The fibrous layer forces the wave through the AVN, which delays it; the bundle of His and Purkyne fibres make the ventricles contract from the apex upwards.

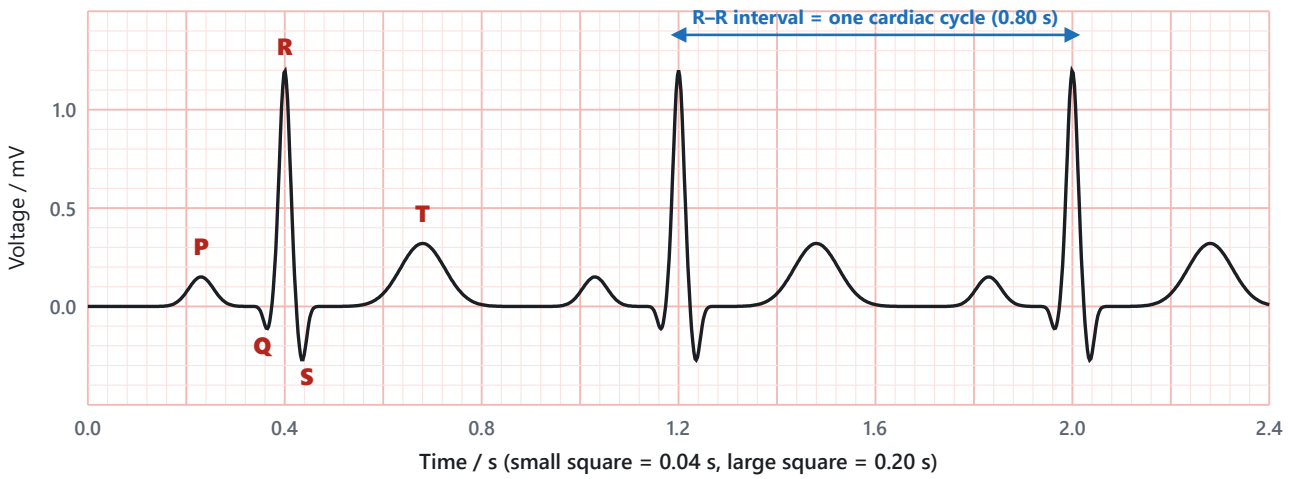
The second interactive diagram at the foot of the page follows the pressures and valves on the left side of the heart through one beat.

The ECG

An **electrocardiogram** records the electrical activity of the heart using electrodes on the skin.

Feature	What's happening
P wave	depolarisation of the **atria** → atrial systole
PR interval	time for the impulse to cross the atria and the AVN delay
QRS complex	depolarisation of the **ventricles** → ventricular systole (the biggest wave: ventricles have the most muscle)
T wave	**repolarisation** of the ventricles → ventricular diastole begins
R–R interval	one complete cardiac cycle

(The atria repolarise during the QRS complex, so this doesn't show as a separate wave.)



A normal ECG. Each large square is 0.20 s. The R–R interval is 0.80 s, so the heart rate is 60 ÷ 0.80 = 75 beats per minute.

Heart rate from an ECG

- **Heart rate (beats min^{-1}) = 60 ÷ time for one cycle in seconds**, measured from one R peak to the next (or between any two matching points).
- Or count beats: heart rate = number of beats × 60 ÷ time in seconds.

WORKED EXAMPLE

Worked example: heart rate from an ECG

In the figure, R to R = four large squares = 4 × 0.20 = 0.80 s. Heart rate = 60 ÷ 0.80 = **75 beats min^{-1}**.

Counting instead: 19 beats in 7.0 s → 19 × 60 ÷ 7 = 162.8 = **163 beats min^{-1}** (WBI15 Jun 2021). Heart rate is a whole number, never a fraction.

From paper speed: at 25 mm s^{-1}, an R–R distance of 20 mm is 20 ÷ 25 = 0.80 s → 75 beats min^{-1}.

Heart rate from an ECG

1. If the P-to-P interval is 0.8 s, one beat takes 0.8 s.
2. Beats per minute = 60 ÷ 0.8 = 75 bpm.

Dividing 60 by the time for one beat converts it to beats per minute.

Abnormal rhythms

Rhythm	What the ECG shows
Tachycardia	fast resting rate (above about 100 beats min^{-1}): R peaks close together
Bradycardia	slow resting rate (below about 60 beats min^{-1}): R peaks far apart. Normal in many trained athletes
Atrial fibrillation	no clear P waves, an irregular wavy baseline, irregular spacing of QRS complexes
Ventricular fibrillation	a chaotic trace with no recognisable P, QRS or T: the ventricles quiver and don't pump (needs a defibrillator)
Ectopic beat	an extra, early beat followed by a longer gap
Heart block	a long PR interval, or P waves not followed by a QRS complex (the AVN delays or blocks the impulse)

- Doctors compare a patient's trace with a normal trace: the shape, height and duration of each wave, the intervals between them, the rate and the regularity. ECGs can also show heart muscle damaged by a heart attack (for example a changed S–T segment or T wave).

MARK-SCHEME WORDING

- Myogenic: contraction "**initiated within the muscle** itself", "without nervous stimulation".
- Comparing ECGs: a comparison in the **same sentence**, using the labels P, QRS, T, R: "P wave smaller in A", "R peak sharper", "QRS duration shorter", "T wave larger", "heart rate lower in the normal trace", "shorter T–P gap in the patient". Converse accepted. Unlabelled descriptions gain nothing.
- Heart rate: show the count and the time (19 beats in 7 s → 163 bpm); a whole number.
- Impulse route: SAN → AVN → bundle of His → Purkyne fibres; the "**delay** at the AVN **allows the atria to empty**" (ventricles to fill).
- Designing an ECG study: record ECGs **before and after** the stimulus (or in two matched groups) and compare heart rate or the interval between the same peaks.

Often asked as: define myogenic; label the SAN and AVN; calculate heart rate from an ECG; compare two ECG traces; name the stage of the cycle at T; design an ECG investigation (posture, energy drinks).

COMMON MISTAKE

- "The ventricles contract from the top (base) downwards." They contract **from the apex upwards**, so blood is pushed up towards the arteries.
- Saying the cardiovascular centre or a nerve **starts** each heartbeat. The SAN starts it; nerves only change the rate.

- Describing trace A, then trace B, without comparing them point by point.
- Measuring between different points on each beat, or giving a heart rate as a decimal.
- Not knowing that the **T wave** marks ventricular repolarisation (the start of ventricular **diastole**).

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- Named arrhythmias haven't been required: abnormal rhythm is tested by **comparing traces** (WBI15 Jun 2021, Jan 2022, Oct 2023, Jan 2026).
- **POTS** (postural orthostatic tachycardia syndrome): heart rate rises by 30 beats min^{-1} or more, or above 120, within 10 minutes of standing up (WBI15 Jun 2022 article).
- Other measures of cardiovascular health: blood pressure, CRP in the blood, angiograms (WBI15 Jun 2025 article).
- A slower heart rate on an ECG shown as "fewer waves of depolarisation" per minute (WBI15 Oct 2025).

Quick checks

QUICK CHECK

Which structure sets the rate of the heartbeat?

- Atrioventricular node
- Sinoatrial node
- Bundle of His
- Purkinje fibres

The SAN starts each beat and sets its rate. The AVN only delays the signal, and the bundle of His and Purkinje fibres carry it to the ventricles. Choosing the AVN is the usual mistake, because it is the node that gets the most attention in the pathway.

QUICK CHECK

Why must the myosin head detach from actin during the cycle?

- To release calcium back into the sarcoplasmic reticulum
- So that tropomyosin can block the binding site again
- So that a fresh ATP can bind and the head can reattach further along actin
- To shorten the A band

Detachment needs a fresh ATP. Its hydrolysis then returns the head to its upright position, allowing a new attachment further along actin. Calcium is pumped back by ATP, and the A band does not change length during contraction.

QUICK CHECK

A heart beats at 60 beats per minute with a stroke volume of 80 cm3. What is its cardiac output?

- 4800 cm3 min-1
- 140 cm3 min-1
- 480 cm3 min-1
- 1.3 cm3 min-1

Cardiac output = stroke volume × heart rate = 80 × 60 = 4800 cm3 min-1. Adding the two values (140) confuses the formula with a sum, and 480 comes from a slipped digit. Dividing by 60 gives a per-second figure, not per minute.

QUICK CHECK

On an ECG, which wave corresponds to ventricular depolarisation?

- P wave
- QRS complex
- T wave
- PR interval

The QRS complex is ventricular depolarisation, which leads to ventricular systole. The P wave is atrial, the T wave is ventricular repolarisation (relaxation), and the PR interval is a time, not a wave.